THE DIMENSION OF A NOETHERIAN LOCAL RING EQUALS THAT OF ITS COMPLETION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The completion map induces isomorphisms modulo powers of the maximal ideal. Algebra, Lemma \ref{algebra-lemma-completion-complete}, cited at proof lines 10630-10631. Gives the agreement of the quotients by powers of the maximal ideal.
- Lengths can be computed on either side of such an isomorphism. Algebra, Lemma \ref{algebra-lemma-pushdown-module}, cited at proof lines 10632-10634. Makes the lengths of the quotients agree.
- Dimension is characterised by the growth of these lengths. Algebra, Proposition \ref{algebra-proposition-dimension}, cited at proof lines 10636-10637. Concludes equality of dimensions.

Source reference. Original lemma 07NV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9775–9775. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 031C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3979–3979. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02M0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3427–3427. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00KQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 760–760. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00ON: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 901–901. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a Noetherian local ring. Then* $\dim(A) = \dim(A^\wedge)$.

Proof. By Algebra, Lemma 031C the map $A \to A^\wedge$ induces isomorphisms $A/\mathfrak{m}^n = A^\wedge/(\mathfrak{m}^\wedge)^n$ for $n \geq 1$. By Algebra, Lemma 02M0 this implies that

$$\text{length}_A(A/\mathfrak{m}^n) = \text{length}_{A^\wedge}(A^\wedge/(\mathfrak{m}^\wedge)^n)$$

for all $n \geq 1$. Thus $d(A) = d(A^\wedge)$ and we conclude by Algebra, Proposition 00KQ. An alternative proof is to use Algebra, Lemma 00ON. □